Hades Model Report

v0.8.4 GitHub

Hastie & Smith, et al. (2026)

MODEL CONFIGURATION

This report summarises the calibration of the model against the reference dataset, including parameter estimates and convergence diagnostics.

Mode	**Alkaline**
Submode	-
Data suite	**./jbel_saghro.xlsx**
FeO/Fe^{2+}	**0.9**
Stages Run	Primary Primitive Basic

1. Results at a Glance

All of the mineral abundances and fractional crystallisation (FC) values for each phase are shown in the table below, as well as the resultant total fractions for the performed amount of fractional crystallisation over all phases.

The total melt fraction crystallised across all phases is 87.40 wt.%

	Primary	Primitive	Basic	Total
Olivine	18.55	13.3	7.55	15.98
Spinel	16.18	6.89		11.82
Plagioclase		2.91	2.87	1.24
K-feldspar		0.36	2.77	0.29
Biotite		0.02		0.01
Magnetite		0.04	0.02	0.02
Ilmenite	3.29	5.82	3.28	4.23
Titanaugite	61.74	59.51	48.71	60.18
Nepheline	0.25	4.77	30.92	3.65
Nosean/Hauyne		6.15	3.84	2.5
Alkali-amphibole		0.22	0.04	0.08

	Primary	Primitive	Basic	Total
FC%	50.0	65.0	28.0	

Table 1: Optimised mineral weights for the model phases (wt.%). Blank rows indicate minerals that were not used in the optimisation for this phase.

2. Parameter Configuration

The parameters below define the model as it was executed for this run.

	Primary	Primitive	Basic	Intermediate	Evolved
Plagioclase	1.2	1.2	1.2	1.2	1.2
Clinopyroxene	0.28	0.28	0.28	0.28	0.28
Orthopyroxene	0.29	0.29	0.29	0.29	0.29
Amphibole	0.37	0.37	0.37	0.37	0.37
K-feldspar	1.3	1.3	1.3	1.3	1.3
Biotite	1.0	1.0	1.0	1.0	1.0
Titanaugite	1.0	1.0	1.0	1.0	1.0
Nepheline	1.0	1.0	1.0	1.0	1.0
Leucite	1.0	1.0	1.0	1.0	1.0
Nosean/Hauyne	1.0	1.0	1.0	1.0	1.0
Alkali-amphibole	1.0	1.0	1.0	1.0	1.0

Table 2: Partition coefficients for the model phases.

3. Magma Evolution

Initial Compositions

	Sample	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
Primary	TL6	40.32	3.03	11.2	12.76	13.62	14.83	2.89	1.35
Primitive	TAB3	44.17	2.89	14.06	10.99	8.68	11.35	4.6	3.25
Basic	TG1	53.29	0.92	20.18	5.33	1.28	3.15	9.62	6.23
Intermediate	TAB7	56.6	0.33	21.2	3.74	0.16	1.14	9.6	7.23
Evolved									
End									

Table 3: Major oxide compositions of the model phases (wt.%).

Computed Compositions

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
Primary	40.32	3.03	11.2	12.76	13.62	14.83	2.89	1.35
Primitive	44.16	2.89	13.17	11.8	8.67	11.33	5.29	2.69
Basic	52.78	0.92	19.3	5.09	1.29	3.24	10.89	6.5
Intermediate	55.45	0.33	21.19	1.19	0.16	1.29	12.11	8.27
Evolved								

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
End								

Table 4: Major oxide compositions of the model phases (wt.%).

4. Optimisation Results

Primary > Primitive

PRIMARY

Population size	50
Iterations	500/500
Cost function	1.13
Trace analysis?	False

Magma Inspection

	Initial	Expected	Computed	% Error
SiO2	40.32	44.17	44.16	0.03
TiO2	3.03	2.89	2.89	0.24
Al2O3	11.2	14.06	13.17	6.36
Fe2O3	12.76	10.99	11.8	7.31
MgO	13.62	8.68	8.67	0.1
CaO	14.83	11.35	11.33	0.13
Na2O	2.89	4.6	5.29	15.14
K2O	1.35	3.25	2.69	17.2

Table 5: Magma compositions for this phase (wt.% for oxides, ppm for trace elements).

Optimised Parameters

	Value
FC%	50.0
Olivine	18.55
Spinel	16.18
Plagioclase	0.0
K-feldspar	0.0
Biotite	0.0
Magnetite	0.0
Ilmenite	3.29
Titanaugite	61.74
Nepheline	0.25
Leucite	0.0
Nosean/Hauyne	0.0
Alkali-amphibole	0.0

Table 6: Optimised parameters for this phase. Fractional crystallisation (FC) is expressed as a percentage, and mineral weights are wt.%.

Bulk Crystallising Assemblage Composition

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
wt.%	36.44	3.17	9.23	13.9	18.43	18.32	0.49	0.01

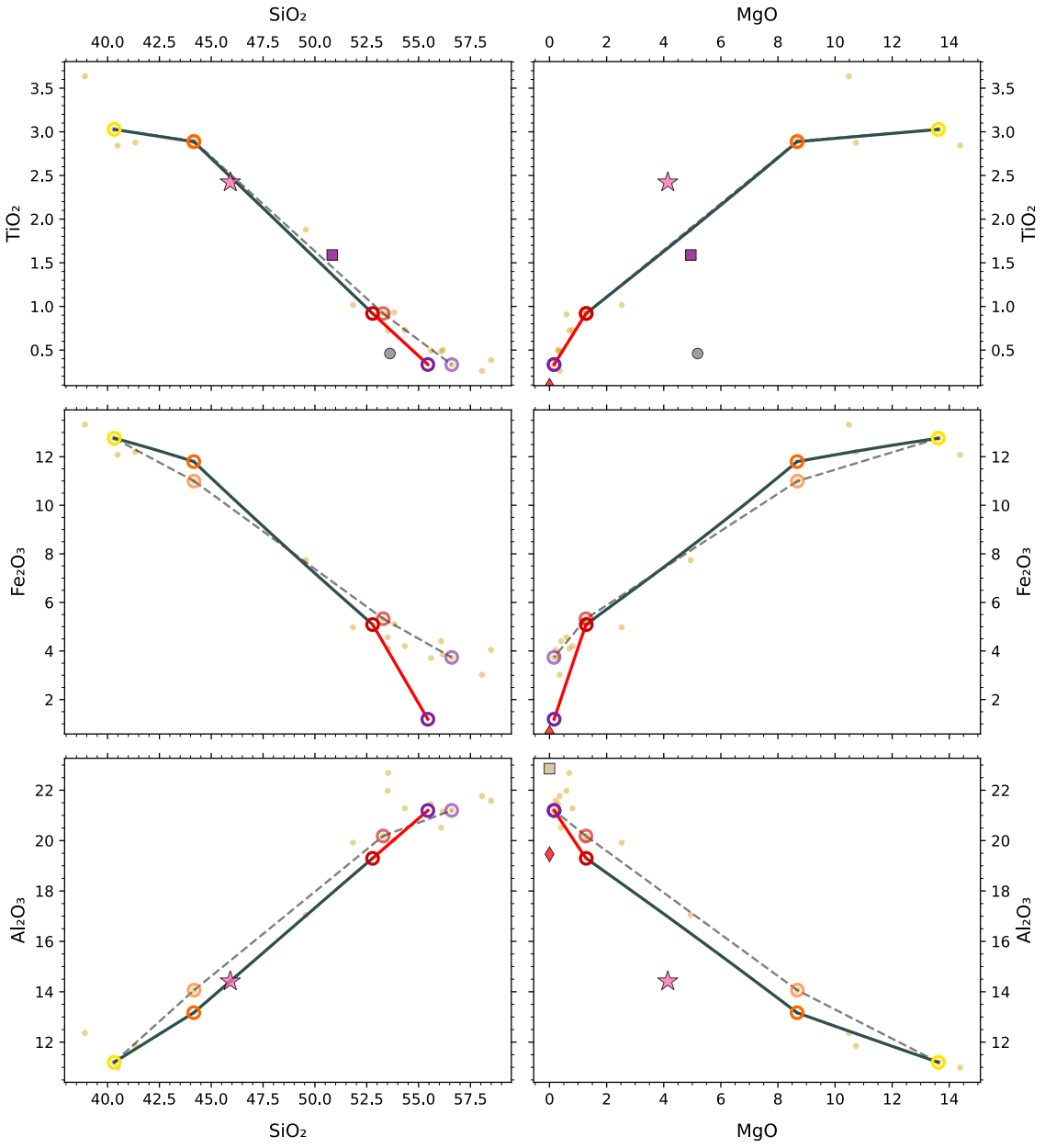
Table 7: Bulk mineralogy of the model phases (wt.%).

Cotectic Compositions

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
olivine	39.47	0.09	0.08	14.49	45.52	0.35	0.0	0.0
spinel	0.26	0.53	30.33	30.52	13.68	24.34	0.34	0.0
plagioclase								
kfeldspar								
biotite								
magnetite								
ilmenite	0.0	50.0	0.0	50.0	0.0	0.0	0.0	0.0
titanaugite	46.93	2.3	6.84	7.51	12.59	23.18	0.66	0.0
nepheline	43.7	0.0	34.78	0.0	0.05	4.46	12.13	4.89
leucite								
nosean_hauyne								
alkali_amphibole								

Table 8: Cotectic compositions of the model phases (wt.%). Blank rows indicate minerals that were not used in the optimisation for this phase.

Bivariate Plot



Bivariate plots continued overleaf.

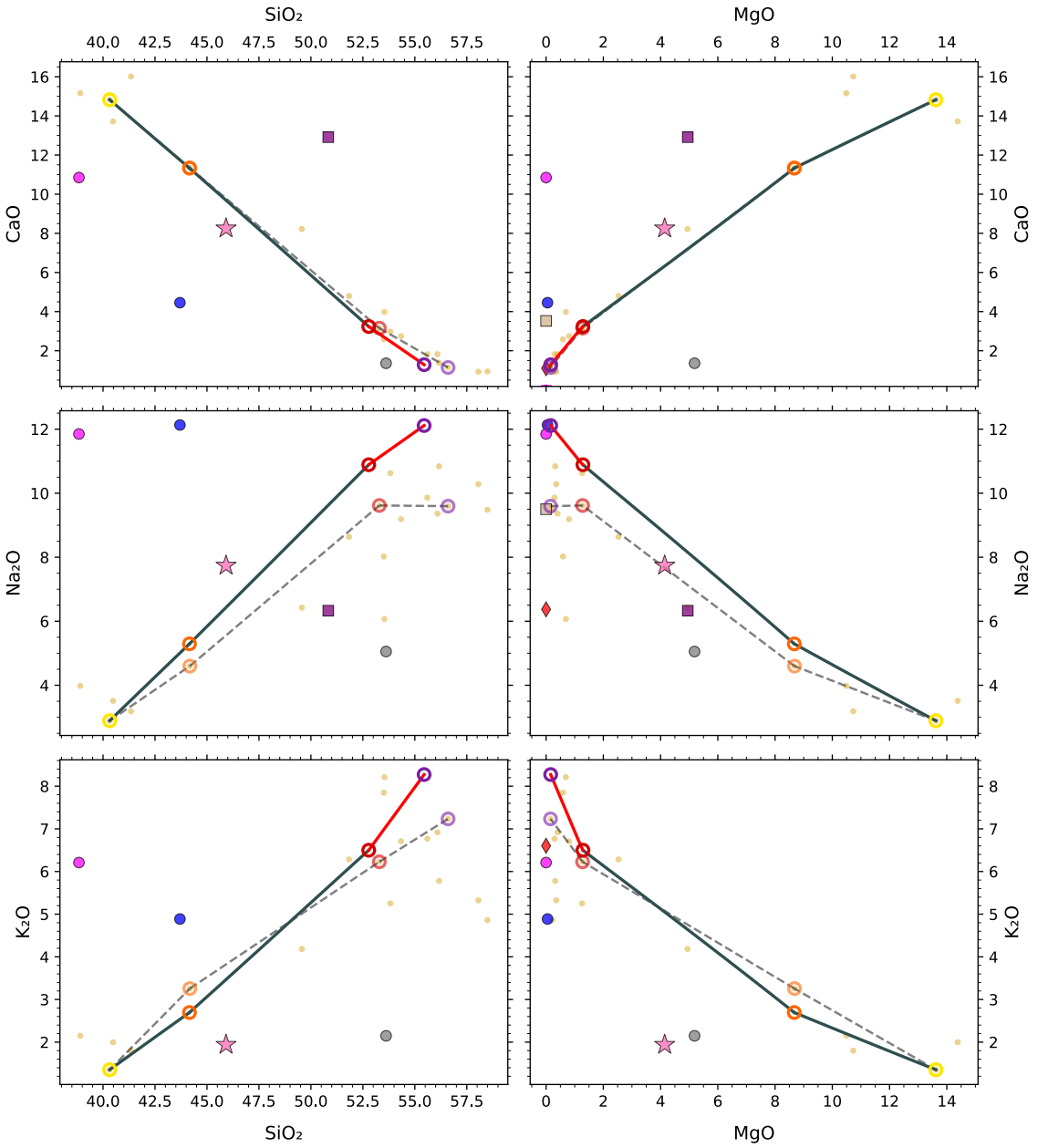


Figure 1: Bivariate plots of the data suite with the data-based and modelled LLODs. The Primary to Primitive transition is highlighted in red and mineral compositions are also displayed for this phase transition. See the legend at the end of the PDF report for the legend.

Primitive > Basic

PRIMITIVE

Population size	50
Iterations	500/500
Cost function	0.95
Trace analysis?	False

Magma Inspection

	Initial	Expected	Computed	% Error
SiO2	44.16	53.29	52.78	0.96
TiO2	2.89	0.92	0.92	0.04

	Initial	Expected	Computed	% Error
Al2O3	13.17	20.18	19.3	4.4
Fe2O3	11.8	5.33	5.09	4.53
MgO	8.67	1.28	1.29	0.9
CaO	11.33	3.15	3.24	2.63
Na2O	5.29	9.62	10.89	13.24
K2O	2.69	6.23	6.5	4.36

Table 9: Magma compositions for this phase (wt.% for oxides, ppm for trace elements).

Optimised Parameters

	Value
FC%	65.0
Olivine	13.3
Spinel	6.89
Plagioclase	2.91
K-feldspar	0.36
Biotite	0.02
Magnetite	0.04
Ilmenite	5.82
Titanaugite	59.51
Nepheline	4.77
Leucite	0.0
Nosean/Hauyne	6.15
Alkali-amphibole	0.22

Table 10: Optimised parameters for this phase. Fractional crystallisation (FC) is expressed as a percentage, and mineral weights are wt.%.

Bulk Crystallising Assemblage Composition

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
wt.%	39.47	3.94	10.0	15.24	12.69	15.72	2.28	0.65

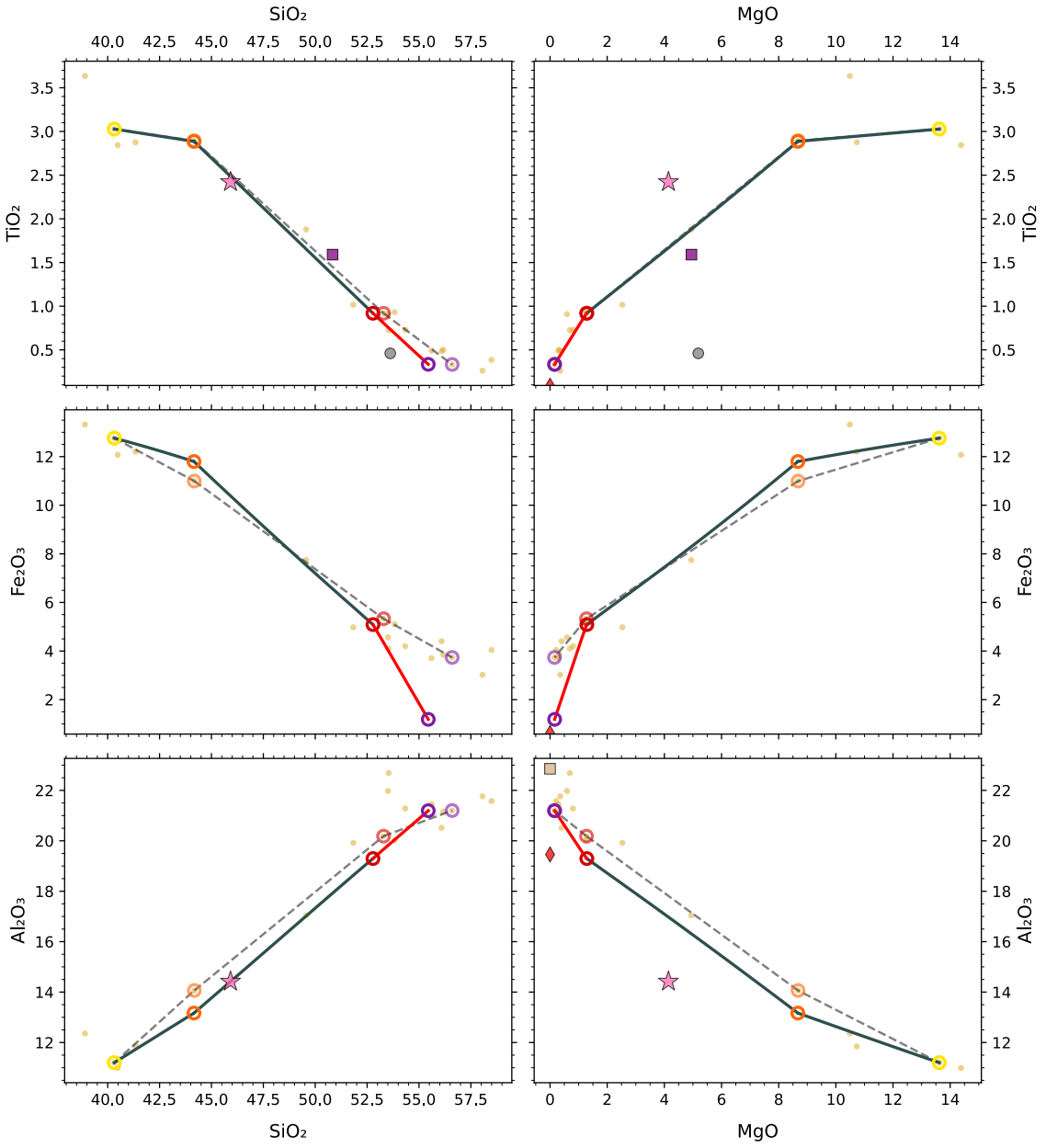
Table 11: Bulk mineralogy of the model phases (wt.%).

Cotectic Compositions

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
olivine	38.95	0.03	1.19	13.95	45.25	0.63	0.0	0.0
spinel	0.26	0.53	30.33	30.52	13.68	24.34	0.34	0.0
plagioclase	53.51	0.0	29.78	0.05	0.14	11.89	4.45	0.18
kfeldspar	65.61	0.0	20.82	0.0	0.0	0.5	7.58	5.49
biotite	39.01	3.02	15.02	19.16	13.73	1.49	0.51	8.06
magnetite	0.0	5.0	0.0	95.0	0.0	0.0	0.0	0.0
ilmenite	0.0	50.0	0.0	50.0	0.0	0.0	0.0	0.0
titanaugite	46.85	1.65	5.32	13.87	9.6	21.39	1.31	0.0
nepheline	43.7	0.0	34.78	0.0	0.05	4.46	12.13	4.89
leucite								
nosean_hauyne	38.84	0.0	32.25	0.0	0.0	10.85	11.85	6.21
alkali_amphibole	53.62	0.46	0.62	31.55	5.19	1.36	5.05	2.15

Table 12: Cotectic compositions of the model phases (wt.%). Blank rows indicate minerals that were not used in the optimisation for this phase.

Bivariate Plot



Bivariate plots continued overleaf.

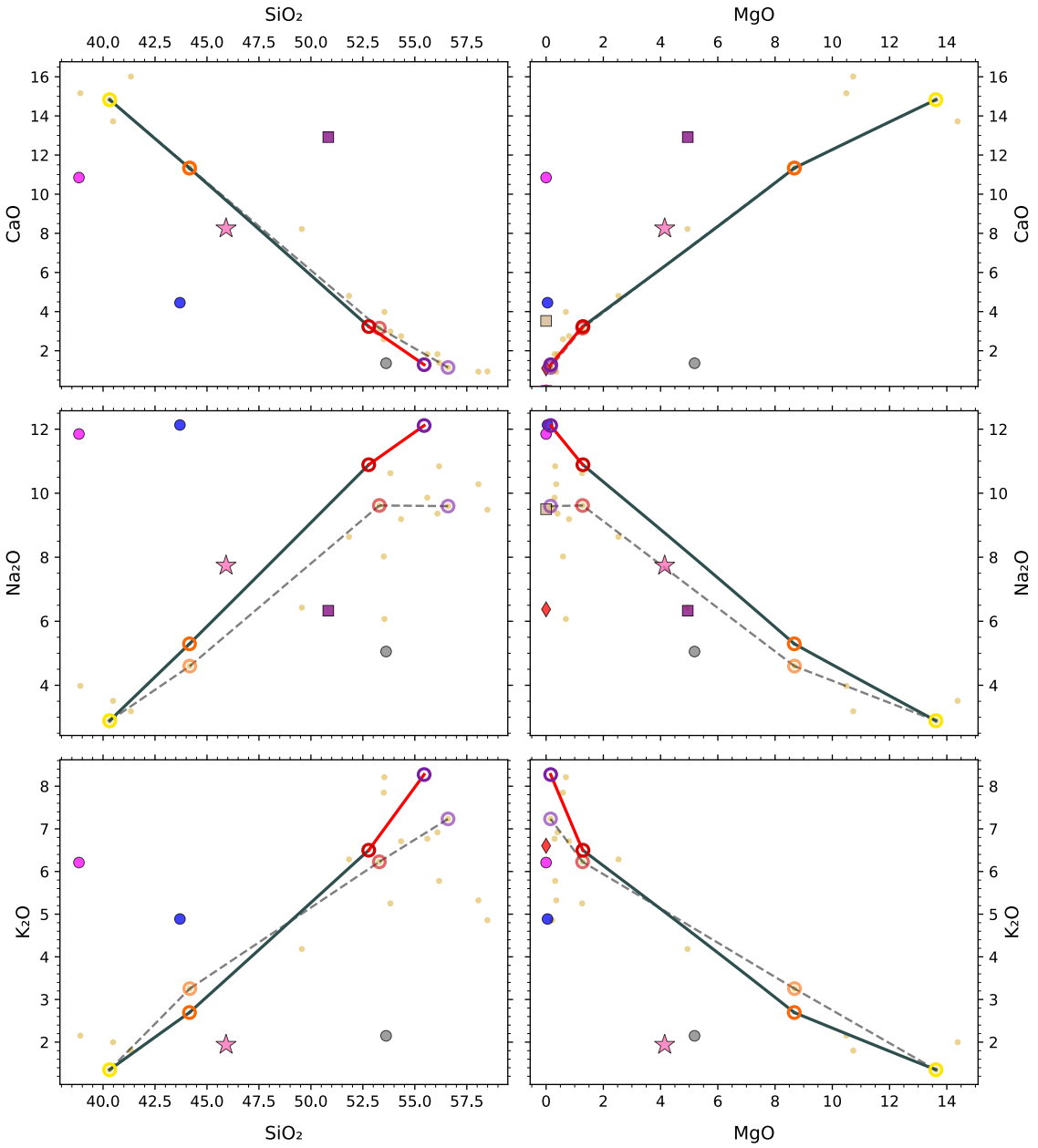


Figure 2: Bivariate plots of the data suite with the data-based and modelled LLODs. The Primitive to Basic transition is highlighted in red and mineral compositions are also displayed for this phase transition. See the legend at the end of the PDF report for the legend.

Basic > Intermediate

BASIC

Population size	50
Iterations	500/500
Cost function	2.82
Trace analysis?	False

Magma Inspection

	Initial	Expected	Computed	% Error
SiO2	52.78	56.6	55.45	2.04
TiO2	0.92	0.33	0.33	0.51

	Initial	Expected	Computed	% Error
Al2O3	19.3	21.2	21.19	0.03
Fe2O3	5.09	3.74	1.19	68.16
MgO	1.29	0.16	0.16	3.0
CaO	3.24	1.14	1.29	12.38
Na2O	10.89	9.6	12.11	26.24
K2O	6.5	7.23	8.27	14.36

Table 13: Magma compositions for this phase (wt.% for oxides, ppm for trace elements).

Optimised Parameters

	Value
FC%	28.0
Olivine	7.55
Spinel	0.0
Plagioclase	2.87
K-feldspar	2.77
Biotite	0.0
Magnetite	0.02
Ilmenite	3.28
Titanaugite	48.71
Nepheline	30.92
Leucite	0.0
Nosean/Hauyne	3.84
Alkali-amphibole	0.04

Table 14: Optimised parameters for this phase. Fractional crystallisation (FC) is expressed as a percentage, and mineral weights are wt.%.

Bulk Crystallising Assemblage Composition

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
wt.%	45.92	2.42	14.42	15.16	4.14	8.25	7.74	1.94

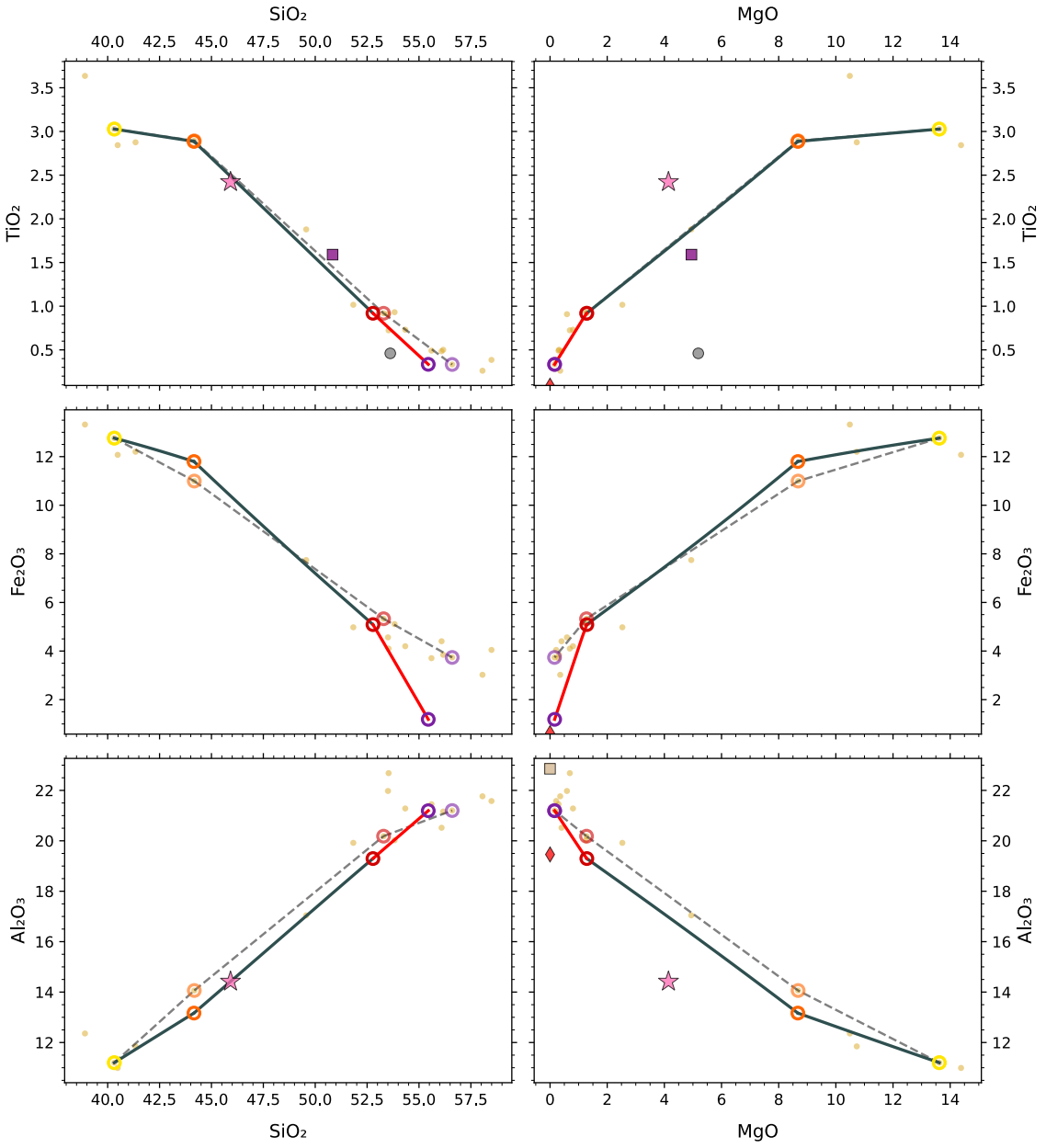
Table 15: Bulk mineralogy of the model phases (wt.%).

Cotectic Compositions

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
olivine	32.85	0.03	1.27	42.64	22.7	0.51	0.0	0.0
spinel								
plagioclase	63.76	0.0	22.86	0.14	0.0	3.52	9.5	0.22
kfeldspar	65.75	0.09	19.46	0.62	0.0	1.1	6.37	6.6
biotite								
magnetite	0.0	5.0	0.0	95.0	0.0	0.0	0.0	0.0
ilmenite	0.0	50.0	0.0	50.0	0.0	0.0	0.0	0.0
titanaugite	50.84	1.59	2.33	21.05	4.95	12.91	6.33	0.0
nepheline	43.7	0.0	34.78	0.0	0.05	4.46	12.13	4.89
leucite								
nosean_hauyne	38.84	0.0	32.25	0.0	0.0	10.85	11.85	6.21
alkali_amphibole	53.62	0.46	0.62	31.55	5.19	1.36	5.05	2.15

Table 16: Cotectic compositions of the model phases (wt.%). Blank rows indicate minerals that were not used in the optimisation for this phase.

Bivariate Plot



Bivariate plots continued overleaf.

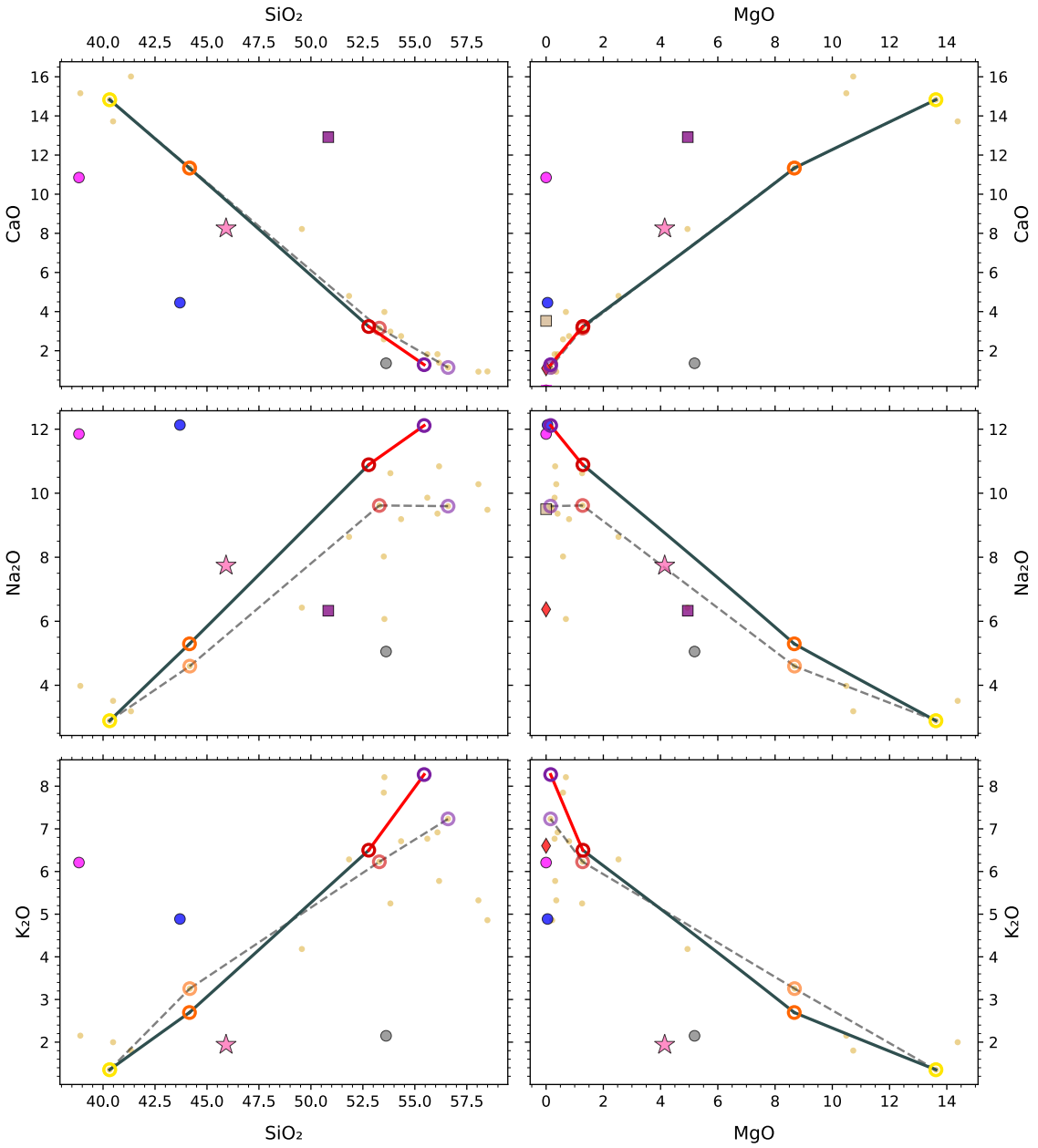


Figure 3: Bivariate plots of the data suite with the data-based and modelled LLODs. The Basic to Intermediate transition is highlighted in red and mineral compositions are also displayed for this phase transition. See the legend at the end of the PDF report for the legend.

5. Figures

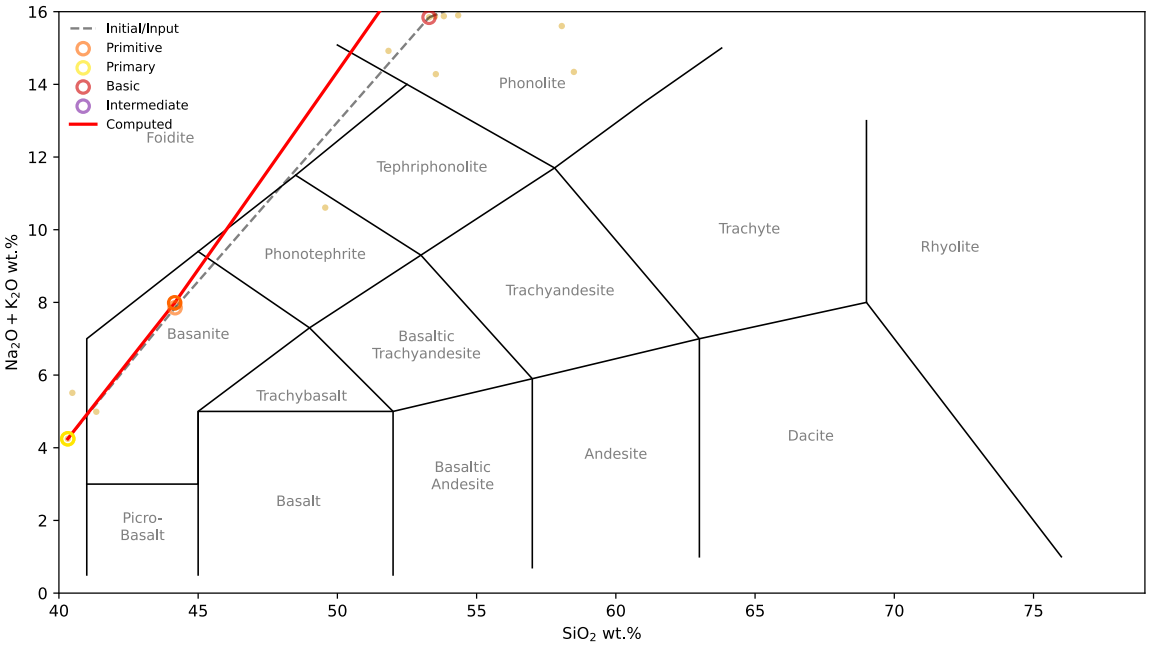


Figure 4: Total Alkali-Silica (TAS) diagram of the model phases.

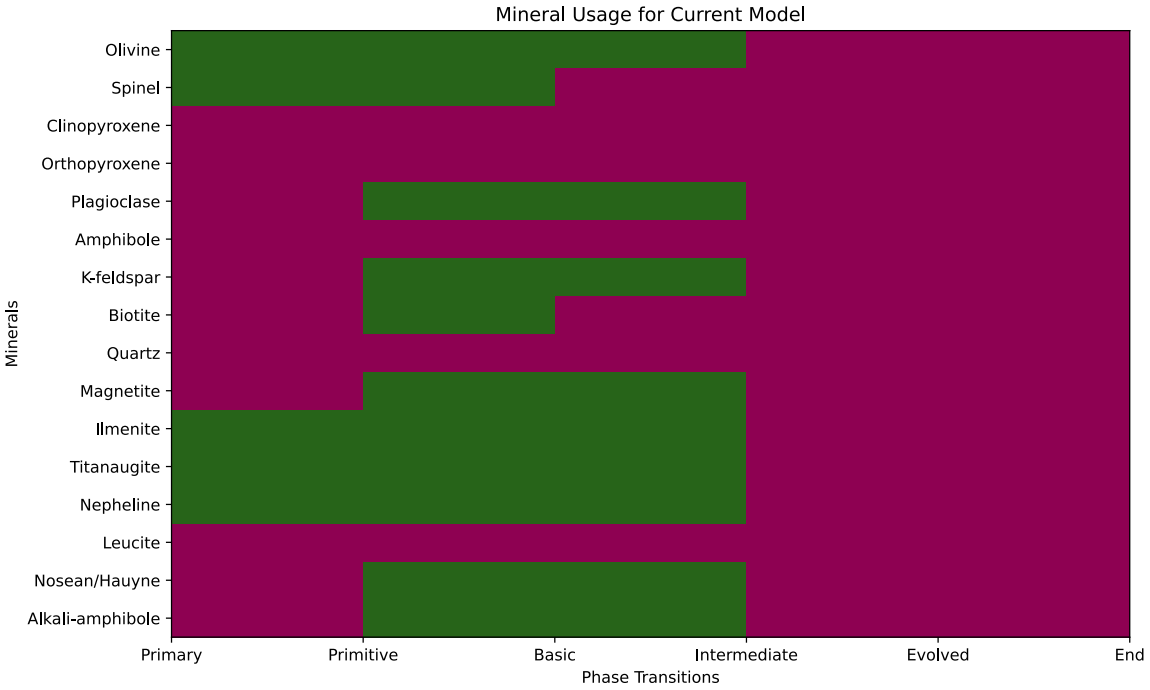


Figure 5: Mineral usage across the model phases.

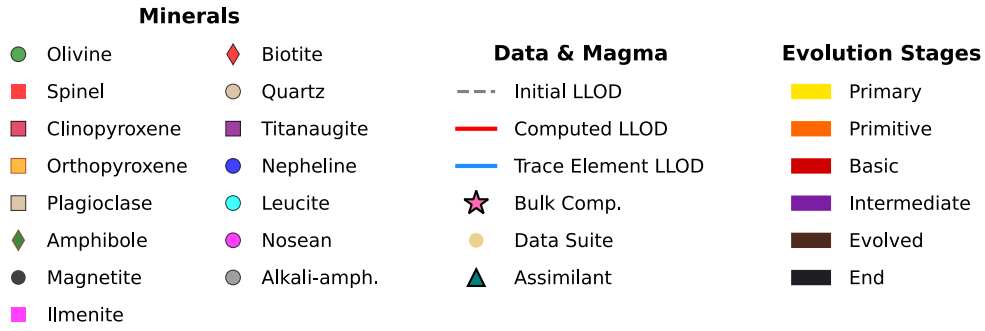


Figure 6: Legend for mineral symbols used in the figures.